

A $4\text{-}\lambda \times 64$ Gb/s Polarization-diverse Silicon Photonic DWDM Receiver using low-PDL 2D Grating Couplers

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Abstract We demonstrate a $4\text{-}\lambda \times 64$ Gb/s silicon-photonic polarization-diverse receiver using a low-PDL 2D grating coupler. Co-optimized grating geometry and fiber coupling angle yield ± 0.2 -dB PDL across a 300-mm wafer. Stable eye diagrams and BER measured under fast 2000-rad/s SOP scrambling confirm fully passive reception independent of polarization.

Introduction

Growing AI/ML workloads demand higher bandwidth density, energy efficiency, and reach from optical interconnects. These demands are accelerating the co-packaging of optical I/O with XPU. Dense wavelength division multiplexing (DWDM) is a key optical I/O architecture to scale throughput [1]. Silicon photonics (SiPh) is especially suited for DWDM links owing to its manufacturability and large-scale integration capability [2–4]. However, silicon waveguide birefringence and single-polarization ring structures make the DWDM receiver (RX) sensitive to the state-of-polarization (SOP) of incoming light. Because the SOP of the optical signal evolves unpredictably as it propagates through standard single-mode fiber (SMF), polarization-dependent reception degrades the link performance. Polarization-maintaining fibre (PMF) mitigates this issue but adds cost and increases insertion loss [5]. Active polarization tracking, typically using thermo-optic phase shifters, must operate continuously and is limited by phase shifters' response [6,7]. Hence, as DWDM scales in wavelength count, passive polarization-diversity schemes are attractive for DWDM RX, avoiding active control while handling arbitrary SOP variations transparently [8–12].

Fig. 1 shows the DWDM transceiver, which employs a microring modulator (MRM) at the transmitter (TX), a microring resonator (MRR) based demultiplexer at the RX, and grating couplers for fiber-to-chip coupling. Grating couplers (GCs) are periodic diffractive structures providing vertical fiber-to-chip coupling at the chip surface. Their advantages — simple fabrication, flexible I/O placement, wafer-level testability and high-density integration — make them a cost-effective SiPh-compatible solution [3]. However, relative to edge couplers, GCs have narrower spectral bandwidth and higher loss, though ongoing research is steadily narrowing both gaps [13–15]. A one-dimensional (1D) GC supports only a single polarization and is therefore used in Fig. 1 to couple light between the laser and the TX, and from the TX output into the SMF. A two-dimensional (2D) GC at the RX couples the transverse electric (TE)

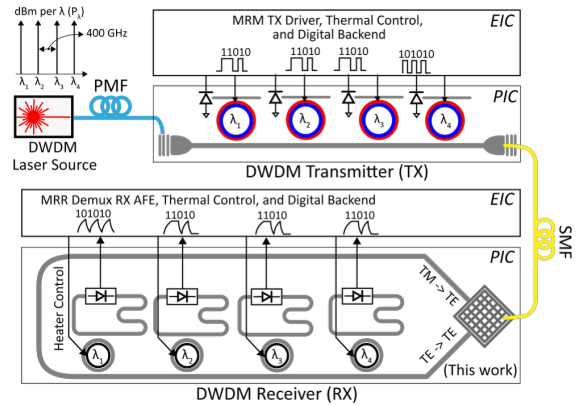


Fig. 1: SiPh DWDM TX-RX link using grating couplers for data transmission over SMF.

and transverse magnetic (TM) polarizations in SMF into on-chip TE modes. Prior passive-diversity receivers fall into two families. The first uses separate MRR banks per polarization, with their drop ports combined into a single dual-input photodetector (PD) [12]; this doubles the thermal-tuning power of the RX, which already dominates the energy budget [2,4]. The second shares a single MRR bank, using a waveguide loop to recombine the two polarization branches [8–10]. The resulting path-length mismatch is compensated by a delay line at the MRR drop port [9], and the approach has been extended to 2D GCs for compact polarization splitting at arbitrary input SOPs [10]. However, because a 2D GC simultaneously performs polarization-sensitive diffraction and polarization splitting, its coupling efficiency is sensitive to modal asymmetry and alignment offsets, increasing polarization-dependent loss (PDL) and degrading link performance [11].

In this work, we reduce the PDL of the 2D GC by co-optimizing the grating geometry and tuning the coupling angle to balance the two orthogonal polarization channels. Combining this low-PDL 2D GC and the MRR-based demultiplexer, we demonstrate a scalable multiwavelength 64-Gb/s channels at 400-GHz spacing, requiring only a single demultiplexer and photodetector per wavelength while eliminating polarization-induced performance penalties.

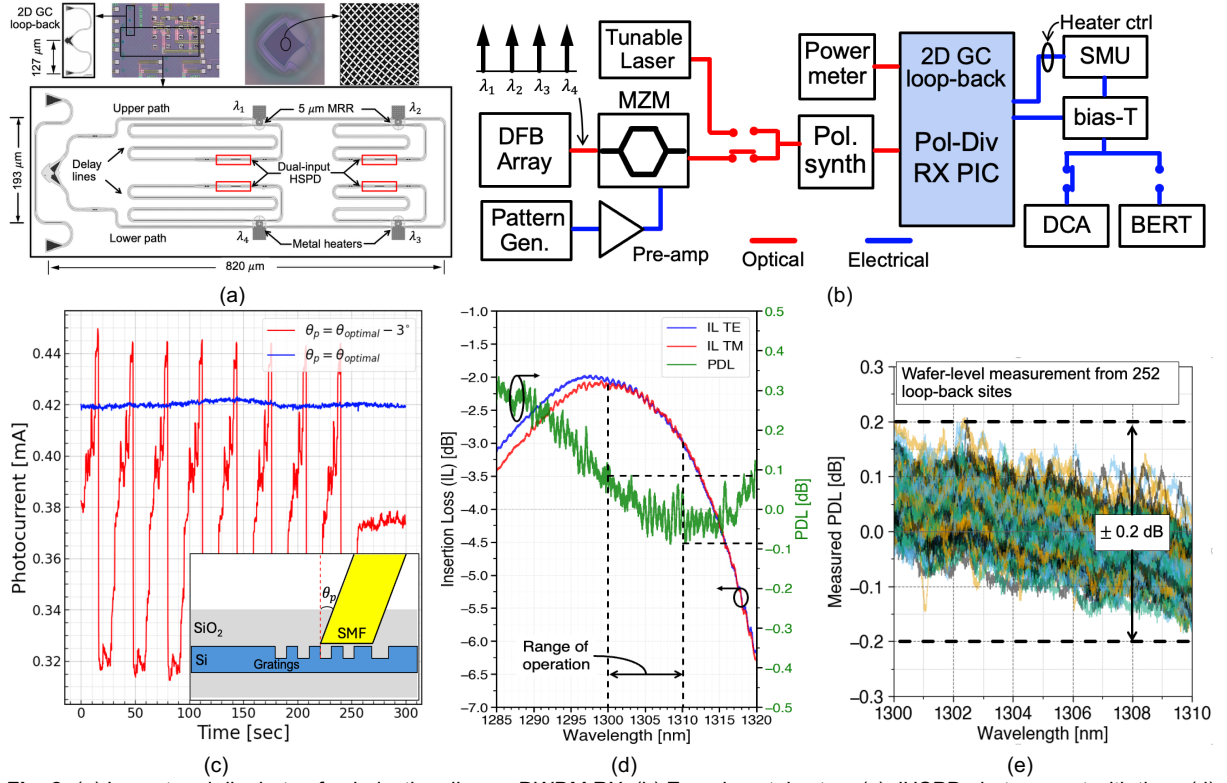


Fig. 2: (a) Layout and diephoto of polarization diverse DWDM RX. (b) Experimental setup. (c) dHSPD photocurrent with time. (d) IL and PDL of 2D GC from a standalone loop-back. (e) Measured PDL from 252 loop-back sites on the same wafer.

Polarization-Diverse DWDM RX Design

Figure 2(a) shows the architecture of the polarization-diverse DWDM RX. The 2D GC decomposes input light of arbitrary SOP into two orthogonal on-chip TE modes routed along an Upper and a Lower path. For the λ_1 channel, Upper-path light reaches a 5-μm-radius MRR directly, while Lower-path light enters the same MRR from the opposite side via a waveguide loop-back. Because the MRR is bidirectionally symmetric, it filters λ_1 from either direction with identical transfer function, and the dropped power from both paths is combined onto a single dual-input high-speed photodetector (dHSPD) which has a 1 A/W responsivity and 3 dB bandwidth of 60 GHz. The compact 5-μm MRR provides sufficient free spectral range to accommodate the four 400-GHz-spaced channels. Carefully designed delay lines compensate the Upper/Lower path mismatch to prevent differential group delay from degrading the eye.

The optimized 2D GC achieves a peak TE insertion loss of 2.0 dB. A 2D GC behaves as two orthogonal 1D gratings sharing a common lattice, each feed waveguide coupling one linear polarization component of the fiber field. The angular dispersion of each polarization's diffracted beam is directed along its own feed axis, so the two polarizations' beam directions coincide on the mirror plane at a single wavelength λ_0 (the zero-PDL wavelength [16]) and separate symmetrically

about the mirror plane away from λ_0 . Two practical consequences follow: the wavelength range of operation should be centered around λ_0 , and the fiber should be tilted at the mirror-axis beam angle $\theta_p(\lambda_0)$.

The DWDM RX is characterized using the test-bench in Fig. 2(b). The optical spectrum occupied by the 400 GHz spaced MRR channels matches that of the PMF-based designs in [2,4]. Figure 2(c) shows fluctuations in dHSPD current with input SOP. These are significantly reduced by optimizing the fiber tilt angle. With combined grating geometry and coupling-angle optimization, the 2D GC achieves $\leq \pm 0.1$ dB PDL across the wavelength range of interest (Fig. 2(d)). The λ_0 is around 1305 nm around which the 10 nm of wavelength range of operation is centred. Further, across 252 loop-back sites on a 300-mm wafer, the PDL spread stays within ± 0.2 dB (Fig. 2(e)), confirming robustness to process variations and suitability for volume production. Figure 3(a) shows the dHSPD photocurrent for the four channels, after the MRRs are thermally tuned. The solid lines show all the power routed through the Upper path and dotted lines show power routed through the lower path. The low PDL between the two paths translates into less than 6% photocurrent variation across all four channels, compared to more than 30% reported in [11]. This reduction in PDL enables DWDM RX performance to become independent of the input SOP.

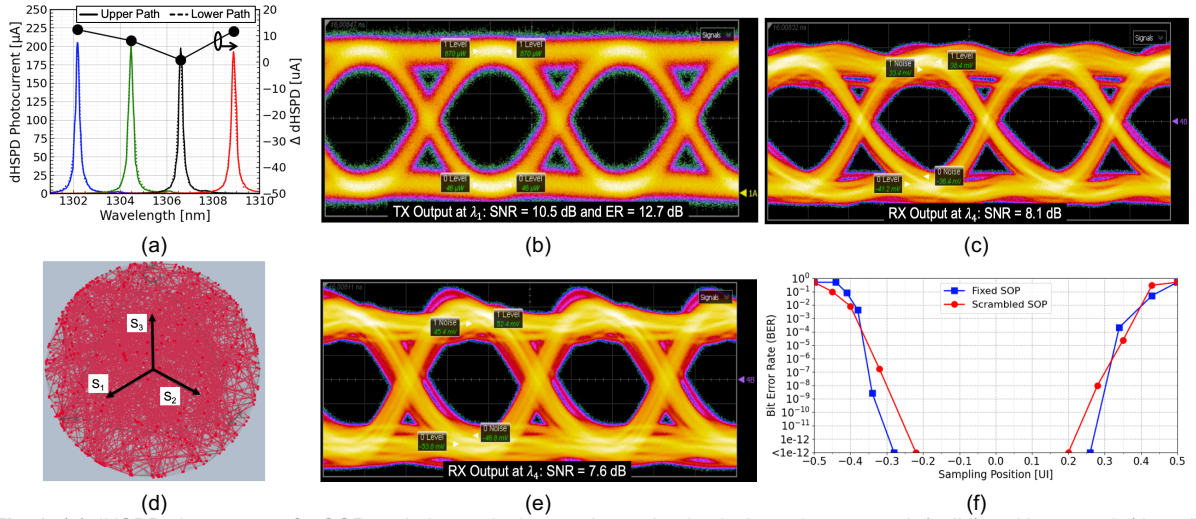


Fig. 3: (a) dHSPD photocurrent for SOP such that optical power is received only through upper path (solid) and lower path (dotted) for each RX channel. (b) TX eye diagram for λ_1 channel at 64 Gb/s. (c) RX eye diagram for λ_4 channel with fixed input SOP. (d) Receiver input SOP trace on the Poincaré sphere after 5 min of random scrambling. (e) RX eye diagram in presence of SOP changing at 2000 rad/s. (f) BER measured in presence of fixed and scrambling SOP. A PRBS15 data pattern is used for this test.

Experimental Results

To verify that the polarization diversity design maintains signal integrity under rapid SOP variations, the test-bench of Fig. 2(b) is used. A DFB array based DWDM laser [17] provides four wavelengths spaced at 400 GHz, which are modulated by the same MZM at 64 Gb/s to load all channels simultaneously. Since the 400 GHz channel spacing is much larger than the 64 Gb/s per-channel bandwidth, the inter-channel cross-talk is negligible. Hence, the shared modulation does not affect the validity of the per-channel BER measurements. Figure 3(b) shows the TX eye at 64 Gb/s for λ_1 captured with the DCA optical head (Fig. 2(b)). Figure 3(c) shows the RX eye for λ_4 channel under fixed input polarization with an arbitrary Upper/Lower power split, recorded with the DCA electrical head. The λ_4 channel is the longest wavelength experiencing highest GC insertion loss and the largest photocurrent variation among the four channels (Fig. 2(e)). Hence, eye at λ_4 is shown in Fig. 3(c) and (e).

Next, a polarization scrambler randomly generates SOP states at a sequence rate of 1.6 kHz on the Poincaré sphere, resulting in an SOP spin rate of approximately 2000 rad/s. An SOP sequence length of $2^{31}-1$ is used, providing sufficient time to measure BER down to the 10^{-12} level while random input SOP is generated. Figure 3(d) shows the accumulated SOP trajectory on the Poincaré sphere over a 5-minute measurement window, confirming that the sphere is uniformly and thoroughly traversed. Figure 3(e) shows the RX eye diagram at 64 Gb/s for λ_4 channel in the presence of rapid SOP scrambling. Comparing the eye diagrams of Fig. 3(c) (fixed SOP) and Fig. 3(e) (scrambled SOP), the SNR degrades by only 0.4 dB. No significant horizontal shift is observed between the two eyes, indicating that the delay-

line compensation maintains stable timing across all polarization states. This is further substantiated by the bathtub curves measured using a BERT, shown in Fig. 3(f). The bathtub opening with SOP scrambling is only marginally smaller than without, and the slight narrowing can be attributed to the residual PDL of the DWDM RX combined with additional PDL contributed by the test-bench components, estimated at approximately 0.2 dB. Notably, the tested SOP spin rate of ~2000 rad/s far exceeds the polarization change rates of 250 rad/s encountered in practical deployed fiber links, providing substantial design margin. The eye diagrams and bathtub curves confirm that the low PDL of the 2D GC allows the receiver to withstand fast SOP transients while maintaining signal quality, keeping the receiver architecture simple and avoiding the need for any active polarization control in DWDM IM/DD links.

Conclusions

We have demonstrated a 256 Gb/s ($4\lambda \times 64$ -Gb/s/ λ) silicon-photonic polarization-diverse receiver using low PDL grating couplers. Through combined optimization of the grating design and fibre coupling angle, PDL of $<\pm 0.2$ dB across 252 test sites on a 300-mm wafer is achieved. The receiver delivers robust, fully passive polarization-independent performance across all four channels. The receiver withstands polarization change rates exceeding 2000 rad/s with minimal impact on SNR as confirmed by the transient eye and BER measurement. This confirms the robust operation of the DWDM RX without active polarization control. These results establish a compact, passive, robust, and scalable polarization-diverse WDM receiver architecture suitable for next-generation optical interconnects.

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